

SEN 202 WEEKLY DEV – WEEKEND 1 ASSIGNMENT

SIM – Security Incident Mapping Platform

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Faculty:	Computing
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Course Code:	SEN 202
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Assignment Brief and Tasks

Insecurity is one of the major headaches the present administration and government of Nigeria is having. However, this insecurity is seen to be given birth to by community crime. Studies have shown that campus is one of such communities where community crimes thrive. As a young software engineer, you are charged to contribute to the elimination or reduction of this crime rate by designing a web platform called 'SIM' (Security Incident Mapping) where students/villa mates could quickly report security crimes on campus (e.g. stabbing, raping, stealing/theft, murder etc.). However, due to the sensitivity of the proposed web-based system, you are expected to complete the following tasks:

- a.** Design and send out a questionnaire to the community involved (students, staff, landlords – since students live in their abode – your design should capture this) using GOOGLE FORM.
- b.** Analyze the issue with the Ishikawa problem analysis tool (using DRAW.IO).
- c.** Develop a mini web interface as a solution (avoid complexity because of submission time).
- d.** Sketch use-case diagram(s) to show how your system/app interacts.
- e.** Present a SRS (Software Requirement Specification) document to explain the interactions of major components of your app. The SRS should clearly show the functional and non-functional requirements aspects of your system/software.
- f.** Create a GITHUB account (if you don't have one) or use your existing one.
- g.** Have the app and all its related documentation (e.g. Ishikawa, use-case, SRS document) stored in your repository.
- h.** Send the link to: ajcomponentengineer@gmail.com
- i.** Submission deadline: Friday 10th July, 2026.
- j.** Once your mail is acknowledged, proceed to print out (font size: 10) your SRS, Use-Case, Ishikawa, object listing, and code listing of your project, and insert into your 'on the way' manual for physical submission.

Answers

Answer to Task (a) – Google Form Questionnaire

A pre-design questionnaire was created for distribution via Google Forms, targeting the three community groups affected by campus insecurity: students, staff, and landlords/property managers hosting off-campus students. The form uses a branching first question (“Which best describes you?”) that routes each respondent to a tailored section:

Students: residence, whether they have witnessed/experienced an incident, incident type, whether it was reported and why/why not, willingness to use an anonymous reporting app, and desired features on an incident map.

Staff: role, awareness of recurring issues, highest-risk locations, and usefulness of a centralized reporting/mapping tool for their unit.

Landlords/Property Managers: number of resident students, history of incidents at the property, existing security measures (fencing, lighting, CCTV, guards), and willingness to receive alerts or support tenant use of the app.

The full question set, answer types (multiple choice, checkboxes, linear scale, short answer), and Google Forms setup notes (branching by section, anonymous responses) are documented in **google_form_questionnaire.md** in the project repository.

Answer to Task (b) – Ishikawa (Fishbone) Analysis

The Ishikawa diagram below analyzes the root causes behind the high rate of campus security incidents, grouped into four standard categories: People, Environment, Process, and Technology. This analysis directly motivated SIM’s design – particularly the Process causes (no fast/anonymous reporting channel, slow security response) and Technology causes (no incident mapping or tracking tool, no real-time alerts), which the platform is built to eliminate.

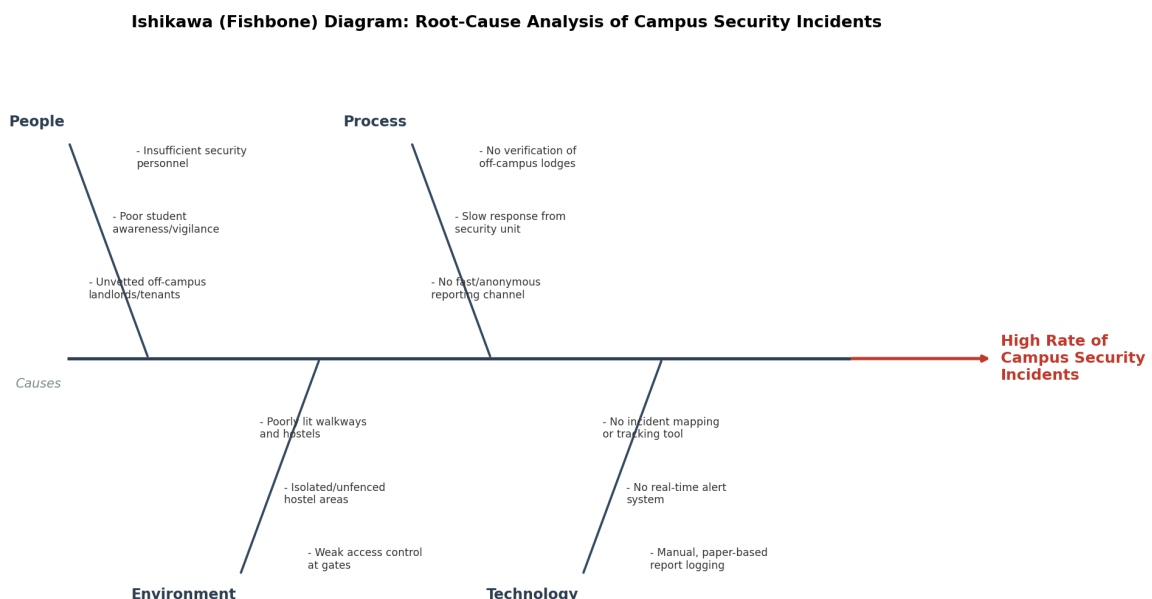
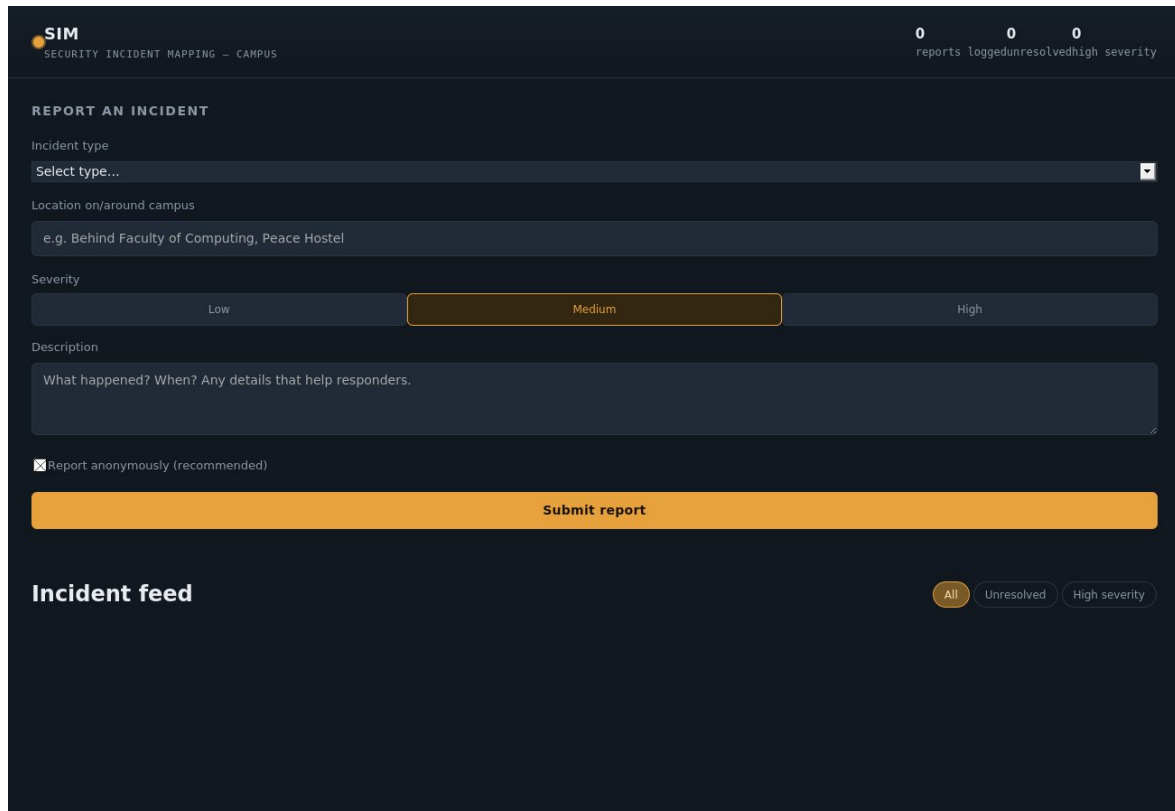


Figure 1: Ishikawa (Fishbone) diagram – root causes of campus security incidents

Answer to Task (c) – Mini Web Interface

A lightweight, single-file web application (HTML, CSS, and vanilla JavaScript) was built to demonstrate the core SIM workflow: reporting an incident, viewing the incident feed, filtering by status/severity, and marking incidents resolved. Data is held in the browser's local storage for this prototype, avoiding the complexity of a full backend given the submission timeline, while still demonstrating a complete, usable interaction flow.



The screenshot displays the SIM (Security Incident Mapping) mini web interface. At the top, the header includes the SIM logo and the text "SECURITY INCIDENT MAPPING – CAMPUS". On the right, a statistics bar shows counts for reports (0), logged in (0), unresolved (0), and high severity (0). The main section is titled "REPORT AN INCIDENT" and contains a form with the following fields: "Incident type" (a dropdown menu with "Select type..." as the placeholder), "Location on/around campus" (a text input field with the example "e.g. Behind Faculty of Computing, Peace Hostel"), "Severity" (three radio buttons for Low, Medium, and High, with Medium selected), and "Description" (a large text area with the placeholder "What happened? When? Any details that help responders."). Below the description field is a checkbox labeled "Report anonymously (recommended)". A prominent orange "Submit report" button is positioned below the form. The bottom section, titled "Incident feed", features three filter chips: "All", "Unresolved", and "High severity".

Figure 2: SIM mini web interface – report form and live incident feed

Key features implemented:

- Report form with incident type, location, severity (low/medium/high), description, and an anonymous-reporting toggle.
- Auto-generated unique reference ID (e.g. SIM-1001) for every submitted report.
- Live incident feed sorted by most recent, with color-coded severity indicators.
- Filter chips for All / Unresolved / High severity.
- One-click status toggle between open and resolved.
- Running statistics strip: total reports, unresolved count, high-severity count.

Object Listing – SIM Web Interface (sim_web_interface.html)

The listing below is the complete source of the mini web interface referenced in Task (c). It is organized into three parts within a single file: CSS styling (visual design), HTML markup (structure: header, report form panel, incident feed panel), and JavaScript logic (state management, rendering, and event handling for the report form, filters, and status toggle).

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SIM &mdash; Security Incident Mapping</title>
<style>
  :root{
    --bg:#12181f;
    --bg-panel:#1a222c;
    --bg-panel-2:#212b37;
    --line:#2b3641;
    --text:#e9edf1;
    --text-dim:#8d99a6;
    --amber:#e8a23d;
    --amber-dim:#7a5a26;
    --red:#c0392b;
    --teal:#3f9c8f;
    --mono: ui-monospace, "SF Mono", "Cascadia Code", Consolas, monospace;
    --sans: -apple-system, "Segoe UI", Roboto, Helvetica, Arial, sans-serif
  ;
  }
  *{box-sizing:border-box;}
  body{
    margin:0; background:var(--bg); color:var(--text);
    font-family:var(--sans); line-height:1.5;
  }
  a{color:var(--amber);}

  header{
    display:flex; align-items:center; justify-content:space-between;
    padding:18px 28px; border-bottom:1px solid var(--line);
    background:linear-gradient(180deg, #151d26, #12181f);
    position:sticky; top:0; z-index:10;
  }
  .brand{display:flex; align-items:center; gap:12px;}
  .brand .dot{
    width:12px; height:12px; border-radius:50%; background:var(--amber);
    box-shadow:0 0 0 0 rgba(232,162,61,.6);
    animation:ping 2.2s infinite;
  }
  @keyframes ping{
    0%{box-shadow:0 0 0 0 rgba(232,162,61,.55);}
    70%{box-shadow:0 0 10px rgba(232,162,61,0);}
    100%{box-shadow:0 0 0 0 rgba(232,162,61,0);}
  }
  .brand h1{font-size:18px; margin:0; letter-spacing:0.5px;}
  .brand .tag{font-size:11px; color:var(--text-dim); font-family:var(--mono
); letter-spacing:1px;}
  .stat-strip{display:flex; gap:22px; font-family:var(--mono); font-size:12
```

```

px; color:var(--text-dim);}
.stat-strip b{color:var(--text); font-size:15px; display:block; font-family:var(--sans);}

main{
  display:grid; grid-template-columns:380px 1fr; gap:0;
  min-height:calc(100vh - 66px);
}
@media (max-width: 860px){ main{grid-template-columns:1fr;} }

.panel{padding:26px; border-right:1px solid var(--line);}
@media (max-width: 860px){ .panel{border-right:none; border-bottom:1px solid var(--line);} }

.panel h2{font-size:14px; text-transform:uppercase; letter-spacing:1.5px; color:var(--text-dim); margin:0 0 18px;}

label{display:block; font-size:12.5px; color:var(--text-dim); margin:14px 0 6px;}
label:first-of-type{margin-top:0;}
select, textarea, input[type=text]{
  width:100%; background:var(--bg-panel-2); border:1px solid var(--line); color:var(--text);
  padding:10px 12px; border-radius:6px; font-family:var(--sans); font-size:14px;
}
select:focus, textarea:focus, input:focus{outline:2px solid var(--amber); outline-offset:1px;}
textarea{resize:vertical; min-height:80px;}

.severity-row{display:flex; gap:8px; margin-top:6px;}
.sev-btn{
  flex:1; text-align:center; padding:8px 4px; border-radius:6px; cursor:pointer;
  border:1px solid var(--line); font-size:12px; color:var(--text-dim); background:var(--bg-panel-2);
}
.sev-btn.active[data-sev="low"]{border-color:var(--teal); color:var(--teal); background:#16302c;}
.sev-btn.active[data-sev="medium"]{border-color:var(--amber); color:var(--amber); background:#332711;}
.sev-btn.active[data-sev="high"]{border-color:var(--red); color:#ff8a7a; background:#331615;}

.anon-row{display:flex; align-items:center; gap:8px; margin-top:14px; font-size:13px; color:var(--text-dim);}

.submit-btn{
  margin-top:20px; width:100%; padding:12px; border:none; border-radius:6px;
  background:var(--amber); color:#1a1305; font-weight:700; font-size:14px; cursor:pointer;
  letter-spacing:0.3px;
}
.submit-btn:hover{background:#f0af57;}

.feed{padding:26px;}

```

```

    .feed-head{display:flex; align-items:center; justify-content:space-between; margin-bottom:16px;}
    .filter-row{display:flex; gap:8px;}
    .filter-chip{
        font-size:12px; padding:6px 12px; border-radius:20px; border:1px solid var(--line);
        color:var(--text-dim); cursor:pointer; background:transparent;
    }
    .filter-chip.active{background:var(--amber-dim); color:#ffe3ba; border-color:var(--amber);}

    .incident{
        border:1px solid var(--line); border-radius:8px; padding:16px 18px; margin-bottom:12px;
        background:var(--bg-panel); position:relative; border-left:3px solid var(--line);
    }
    .incident.sev-low{border-left-color:var(--teal);}
    .incident.sev-medium{border-left-color:var(--amber);}
    .incident.sev-high{border-left-color:var(--red);}

    .incident-top{display:flex; justify-content:space-between; align-items:baseline; gap:10px;}
    .incident-type{font-weight:700; font-size:14.5px;}
    .incident-id{font-family:var(--mono); font-size:11px; color:var(--text-dim);}
    .incident-meta{font-size:12px; color:var(--text-dim); margin-top:4px; font-family:var(--mono);}
    .incident-desc{margin-top:10px; font-size:13.5px; color:#cfd8e0;}
    .incident-foot{margin-top:10px; display:flex; justify-content:space-between; align-items:center;}
    .status-pill{
        font-size:11px; padding:4px 10px; border-radius:20px; font-family:var(--mono);
        border:1px solid var(--line); color:var(--text-dim); cursor:pointer; background:var(--bg-panel-2);
    }
    .status-pill.resolved{color:var(--teal); border-color:var(--teal);}

    .empty{color:var(--text-dim); font-size:13px; text-align:center; padding:40px 0; border:1px dashed var(--line); border-radius:8px;}

    footer{padding:16px 28px; border-top:1px solid var(--line); color:var(--text-dim); font-size:11.5px; font-family:var(--mono);}
</style>
</head>
<body>

<header>
    <div class="brand">
        <span class="dot"></span>
        <div>
            <h1>SIM</h1>
            <div class="tag">SECURITY INCIDENT MAPPING &mdash; CAMPUS</div>
        </div>
    </div>

```

```

<div class="stat-strip">
  <div><b id="statTotal">0</b>reports logged</div>
  <div><b id="statOpen">0</b>unresolved</div>
  <div><b id="statHigh">0</b>high severity</div>
</div>
</header>

<main>
  <section class="panel">
    <h2>Report an incident</h2>
    <form id="reportForm">
      <label for="type">Incident type</label>
      <select id="type" required>
        <option value="">Select type&hellip;</option>
        <option>Theft / Stealing</option>
        <option>Stabbing / Assault</option>
        <option>Rape / Sexual Assault</option>
        <option>Murder</option>
        <option>Burglary (Hostel/Lodge)</option>
        <option>Suspicious Person/Activity</option>
        <option>Other</option>
      </select>

      <label for="location">Location on/around campus</label>
      <input type="text" id="location" placeholder="e.g. Behind Faculty of Computing, Peace Hostel" required>

      <label>Severity</label>
      <div class="severity-row" id="sevRow">
        <div class="sev-btn" data-sev="low">Low</div>
        <div class="sev-btn active" data-sev="medium">Medium</div>
        <div class="sev-btn" data-sev="high">High</div>
      </div>

      <label for="desc">Description</label>
      <textarea id="desc" placeholder="What happened? When? Any details that help responders." required></textarea>

      <div class="anon-row">
        <input type="checkbox" id="anon" checked>
        <label for="anon" style="margin:0;">Report anonymously (recommended)
      </div>

      <button type="submit" class="submit-btn">Submit report</button>
    </form>
  </section>

  <section class="feed">
    <div class="feed-head">
      <h2 style="margin:0;">Incident feed</h2>
      <div class="filter-row" id="filterRow">
        <button class="filter-chip active" data-f="all">All</button>
        <button class="filter-chip" data-f="open">Unresolved</button>
        <button class="filter-chip" data-f="high">High severity</button>
      </div>
    </div>
  </section>

```



```

    </div>
    <div id="incidentList"></div>
  </section>
</main>

<footer>SIM PROTOTYPE &mdash; DEMO DATA STORED LOCALLY IN THIS BROWSER ONLY
. NOT CONNECTED TO CAMPUS SECURITY.</footer>

<script>
  const KEY = "sim_incidents_v1";
  let incidents = JSON.parse(localStorage.getItem(KEY) || "null");

  if (!incidents) {
    incidents = [
      {id:"SIM-1001", type:"Theft / Stealing", location:"Faculty of Computing car park", severity:"medium",
        desc:"Laptop bag stolen from unattended reading room around 2pm.", anon:true, status:"open",
        ts: Date.now() - 1000*60*60*5},
      {id:"SIM-1002", type:"Suspicious Person/Activity", location:"Back gate, near fence", severity:"low",
        desc:"Unknown individual loitering near back fence late at night for two nights running.", anon:true, status:"resolved",
        ts: Date.now() - 1000*60*60*30},
      {id:"SIM-1003", type:"Burglary (Hostel/Lodge)", location:"Peace Lodge, off-campus", severity:"high",
        desc:"Room forced open while occupants were at lectures; valuables missing.", anon:false, status:"open",
        ts: Date.now() - 1000*60*60*2},
    ];
    save();
  }

  let selectedSev = "medium";
  let activeFilter = "all";

  document.getElementById("sevRow").addEventListener("click", e=>{
    const btn = e.target.closest(".sev-btn");
    if(!btn) return;
    document.querySelectorAll(".sev-btn").forEach(b=>b.classList.remove("active"));
    btn.classList.add("active");
    selectedSev = btn.dataset.sev;
  });

  document.getElementById("filterRow").addEventListener("click", e=>{
    const btn = e.target.closest(".filter-chip");
    if(!btn) return;
    document.querySelectorAll(".filter-chip").forEach(b=>b.classList.remove("active"));
    btn.classList.add("active");
    activeFilter = btn.dataset.f;
    render();
  });

  document.getElementById("reportForm").addEventListener("submit", e=>{

```

```

    e.preventDefault();
    const type = document.getElementById("type").value;
    const location = document.getElementById("location").value.trim();
    const desc = document.getElementById("desc").value.trim();
    const anon = document.getElementById("anon").checked;
    if(!type || !location || !desc) return;

    const id = "SIM-" + (1000 + incidents.length + 1);
    incidents.unshift({id, type, location, severity:selectedSev, desc, anon
, status:"open", ts:Date.now()});
    save();
    render();
    e.target.reset();
    document.querySelectorAll(".sev-btn").forEach(b=>b.classList.remove("ac
tive"));
    document.querySelector('.sev-btn[data-sev="medium"]').classList.add("ac
tive");
    selectedSev = "medium";
    document.getElementById("anon").checked = true;
  });

function toggleStatus(id){
  const inc = incidents.find(i=>i.id===id);
  if(!inc) return;
  inc.status = inc.status === "open" ? "resolved" : "open";
  save();
  render();
}

function timeAgo(ts){
  const mins = Math.floor((Date.now()-ts)/60000);
  if(mins < 60) return mins + "m ago";
  const hrs = Math.floor(mins/60);
  if(hrs < 24) return hrs + "h ago";
  return Math.floor(hrs/24) + "d ago";
}

function save(){ localStorage.setItem(KEY, JSON.stringify(incidents)); }

function render(){
  const list = document.getElementById("incidentList");
  let filtered = incidents;
  if(activeFilter === "open") filtered = incidents.filter(i=>i.status===
"open");
  if(activeFilter === "high") filtered = incidents.filter(i=>i.severity==
="high");

  document.getElementById("statTotal").textContent = incidents.length;
  document.getElementById("statOpen").textContent = incidents.filter(i=>i
.status=== "open").length;
  document.getElementById("statHigh").textContent = incidents.filter(i=>i
.severity=== "high").length;

  if(filtered.length === 0){
    list.innerHTML = '<div class="empty">No incidents match this filter.<
/div>';
  }
}

```

```

    return;
}

list.innerHTML = filtered.map(i => `
  <div class="incident sev-${i.severity}">
    <div class="incident-top">
      <span class="incident-type">${i.type}</span>
      <span class="incident-id">${i.id}</span>
    </div>
    <div class="incident-meta">${i.location} &middot; ${timeAgo(i.ts)}
&middot; ${i.anon ? "anonymous" : "identified"} &middot; ${i.severity} seve
rity</div>
    <div class="incident-desc">${i.desc}</div>
    <div class="incident-foot">
      <span class="status-pill ${i.status==='resolved'? 'resolved': ''}"
onclick="toggleStatus('${i.id}')">
        ${i.status === "resolved" ? "&#10003; resolved" : "&#9679; open
&mdash; mark resolved"}
      </span>
    </div>
  </div>
` ).join("");
}

render();
</script>
</body>
</html>

```

Answer to Task (d) – Use-Case Diagram

The use-case diagram below shows how the four actor types – Student, Staff, Landlord/Villa Mate, and Security Admin – interact with the core use cases of the SIM platform. Reporters (students, staff, landlords) can report incidents and view the incident feed; reporting an incident includes selecting an incident type/location and may extend to attaching a description. The Security Admin reviews reports, verifies and updates their status, generates statistics, and receives alert notifications triggered by status updates.

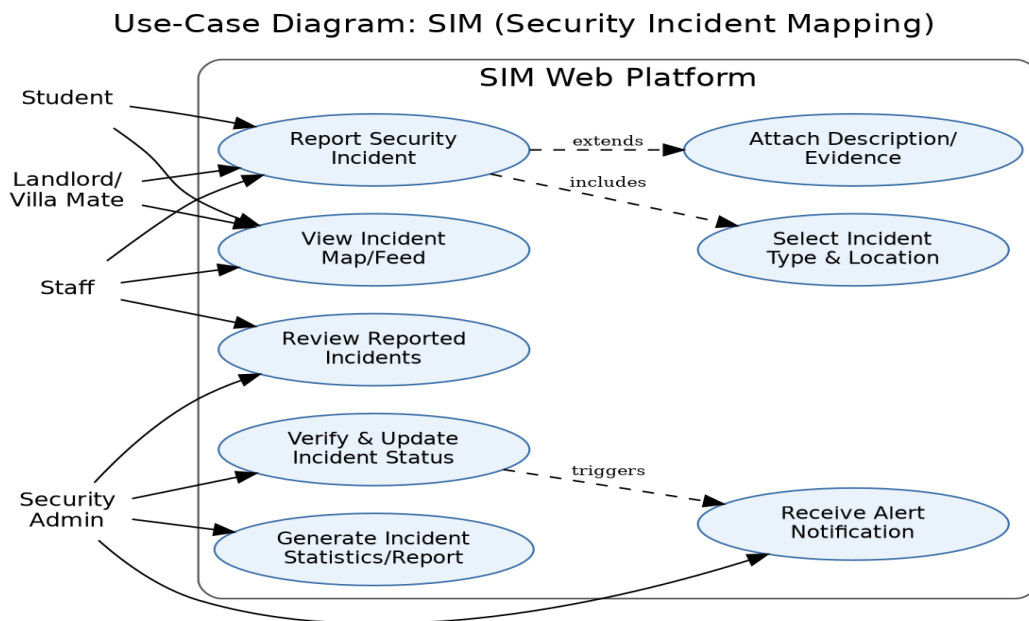


Figure 3: Use-case diagram – SIM actors and their interactions

Answer to Task (e) – SRS (Software Requirements Specification)

A full SRS document (SIM_SRS_Document.docx) was produced covering introduction/scope, user classes, operating environment, functional requirements, non-functional requirements, the use-case diagram, the Ishikawa analysis, and future work. Its functional and non-functional requirements are reproduced below.

Functional Requirements

ID	Requirement	Description
FR1	Submit report	The system shall allow a user to submit an incident report specifying type, location, severity, and description.
FR2	Anonymous reporting	The system shall allow the reporter to mark a report as anonymous, hiding their identity.
FR3	View incident feed	The system shall display a list of submitted incidents ordered by most recent.
FR4	Filter incidents	The system shall allow filtering of the incident feed by status (open/resolved) and by severity (low/high).
FR5	Update status	The system shall allow an authorized user to mark an incident as resolved or reopen it.
FR6	Basic statistics	The system shall display running counts of total reports, unresolved reports, and high-severity reports.
FR7	Unique identification	The system shall assign a unique reference ID to every submitted incident.

Non-Functional Requirements

ID	Requirement	Description
NFR1	Usability	The reporting form shall be simple enough to complete in under 60 seconds on a mobile device.
NFR2	Availability	The web interface shall be accessible whenever a user has an internet connection (target 99% uptime).
NFR3	Privacy	Anonymous reports shall not store or display any personally identifying information.
NFR4	Performance	The incident feed shall render within 2 seconds for up to 500 incidents.
NFR5	Portability	The interface shall render correctly on both desktop and mobile screen sizes (responsive layout).
NFR6	Maintainability	Code shall be modular (separate concerns for form handling, storage, and rendering) to ease future development.

The complete SRS, including the diagrams above and a User Classes table (Reporter and Security Admin), is available in full as its own Word document in the repository for detailed reading and for printing into the physical manual.

Answers to Tasks (f)–(j) – Repository, Submission & Manual

(f) GitHub account/repository: A GitHub repository (e.g. named SEN202 or similarly) is used to store this project, consistent with the existing repository already created for COS202 practical works, or a new dedicated one for SEN202.

(g) Repository contents: The repository holds the mini web interface (sim_web_interface.html), the SRS document (SIM_SRS_Document.docx), the use-case diagram (usecase_diagram.png), the Ishikawa diagram (ishikawa_diagram.png), and the Google Form questionnaire draft (google_form_questionnaire.md).

(h) Submission link: The repository URL is to be sent to ajcomponentengineer@gmail.com.

(i) Deadline: Friday 10th July, 2026.

(j) Physical manual: Once the submission email is acknowledged, this document (containing the SRS content, Use-Case diagram, Ishikawa diagram, and the full object/code listing of the web interface, all set at font size 10) is printed and inserted into the 'on the way' manual as instructed.